

Digital Electronics Lab Report

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Experiment 5: Single-Digit Add/Sub Calculator

Objectives :

1. To design and implement a single-digit calculator capable of performing addition and subtraction operations.
2. To interface a hexadecimal keypad with digital logic circuits for entering numeric data.
3. To display the calculated result on a 7-segment LED display using a BCD-to-7-segment decoder.

Apparatus :

- Hex keypad (1)
- IC 74C922 — Hex keypad encoder (1)

- IC 74374 — Octal D flip-flop (1)
- IC 7486 — Quad XOR gate (1)
- IC 7483 — 4-bit binary adder (1)
- IC 74173 — Quad D flip-flop register (1)
- IC 7447 — BCD to 7-segment decoder (1)
- 7-segment LED display — Common anode type (1)
- Resistors — $470\ \Omega$ / $390\ \Omega$ (8)
- Resistors — $100\ \Omega$ (2)
- Capacitors — $10\ \text{nF}$ (3)
- Capacitor — $100\ \text{nF}$ (1)
- Breadboard
- Connecting wires
- DC power supply — $5\ \text{V}$

Theory :

This experiment implements a single-digit calculator capable of performing addition and subtraction of decimal digits (0–9).

The hexadecimal keypad is used to enter numbers. The IC 74C922 encodes the pressed key into a 4-bit BCD output, which is stored using flip-flop registers (IC 74374 and IC 74173).

Arithmetic operation is performed using IC 7483, a 4-bit binary adder. Subtraction is achieved using the 2's complement method. The IC 7486 (XOR gates) conditionally inverts the second operand when ADD/SUB is HIGH, and a carry input of 1 is added:

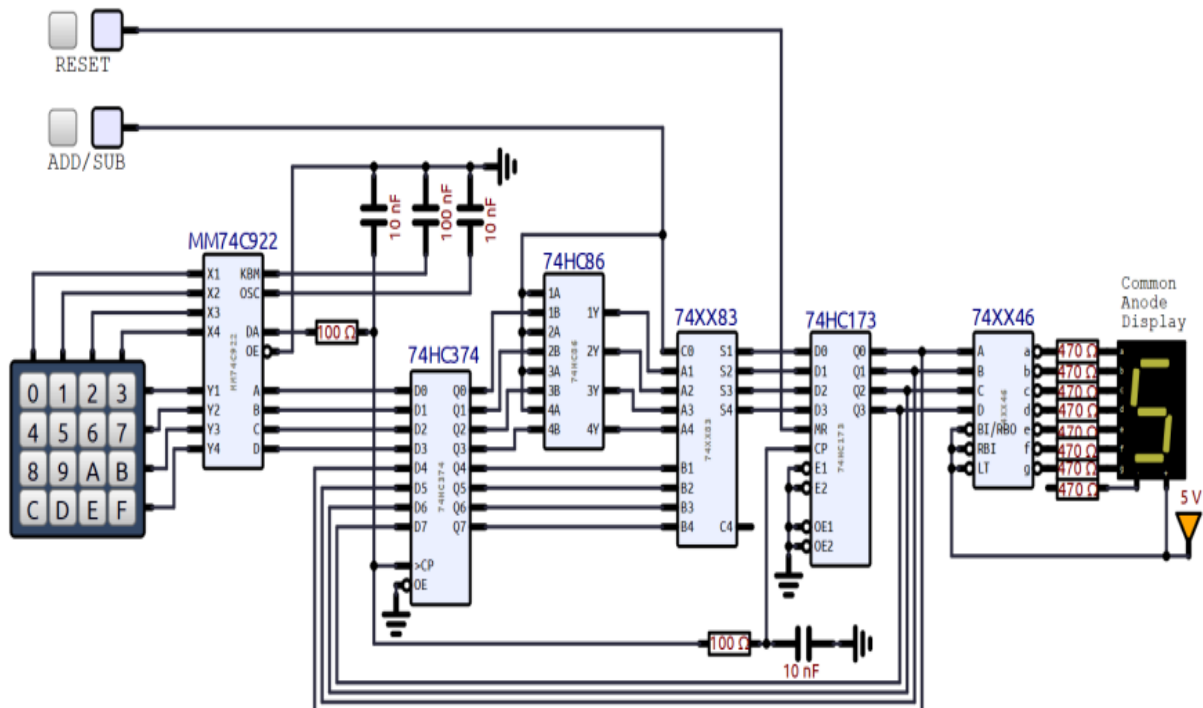
When Add/Sub = Low - addition is performed.

When Add/Sub = High - subtraction is performed.

The result is sent to IC 7447, which drives the common-anode 7-segment display to show the final digit.

Thus, the circuit functions as a basic digital add/sub calculator using combinational and sequential logic components.

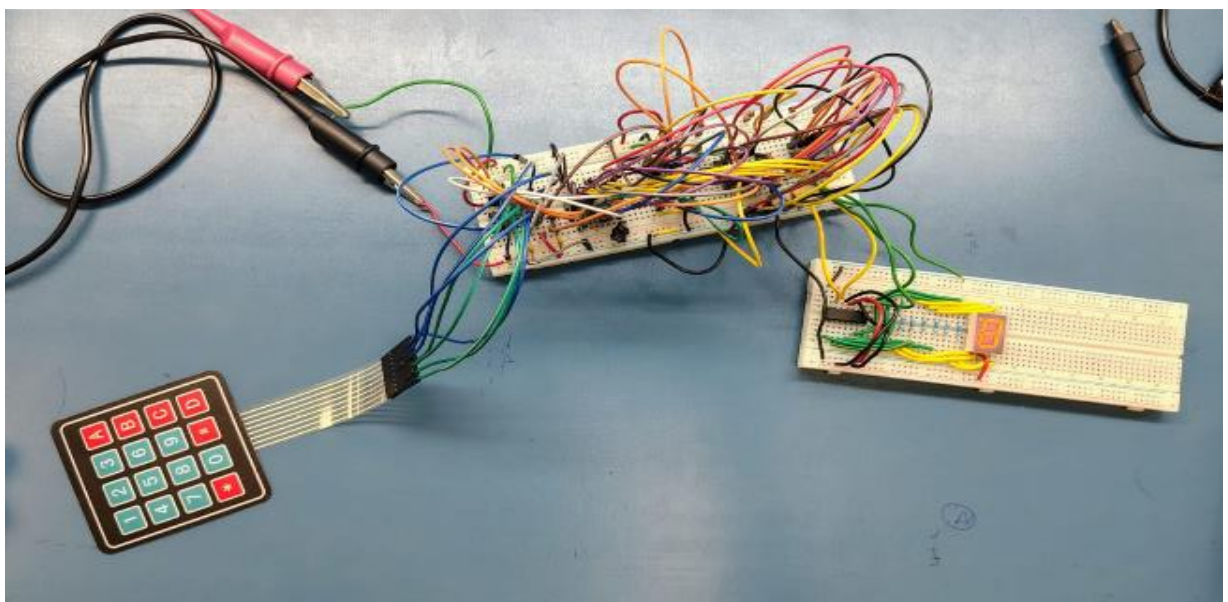
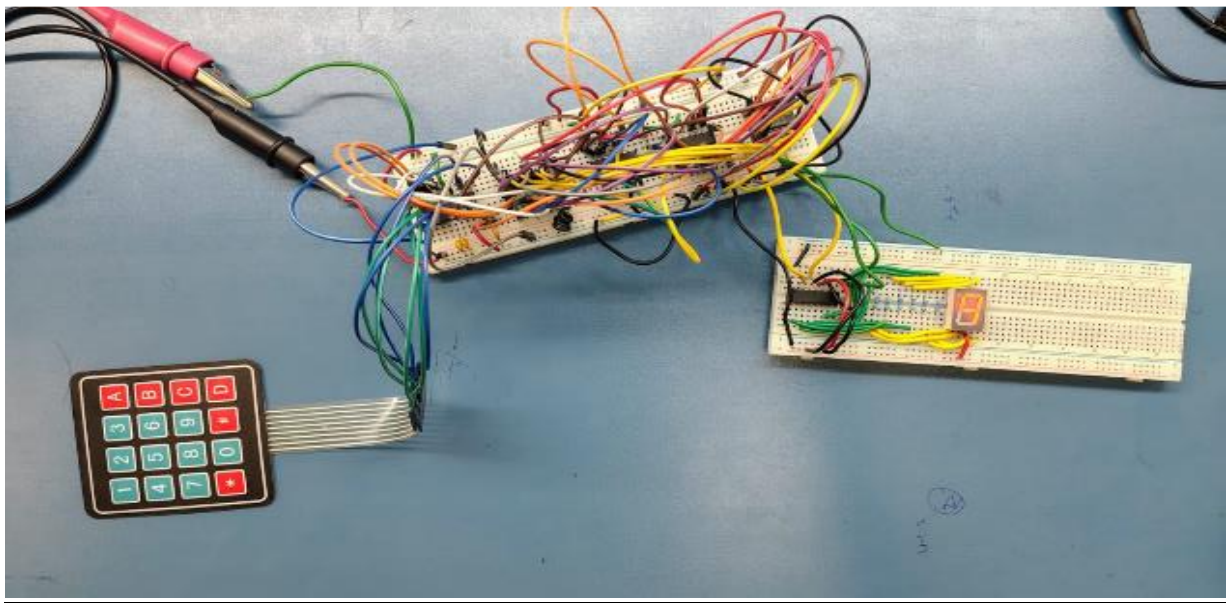
Circuit Diagram :



Observation :

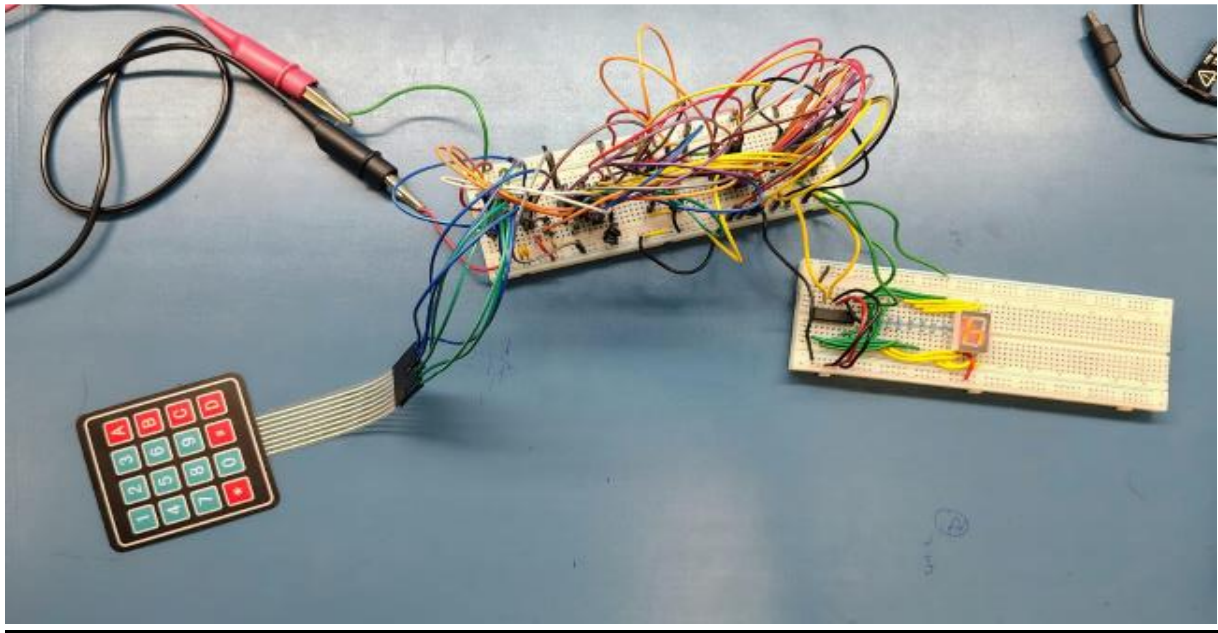
Addition :

6+2=8:



Subtraction :

6-2=4 :



Conclusion :

In this experiment, a single-digit digital calculator was successfully designed and implemented using combinational and sequential logic circuits. The keypad encoder correctly converted key presses into BCD data, and the registers stored the operands and intermediate results. The IC 7483 adder performed arithmetic operations, while the XOR gates enabled subtraction through the 2's complement method. Finally, the BCD-to-7-segment decoder displayed the result on the LED display.

The experiment demonstrated practical implementation of arithmetic logic operations, data storage using flip-flops, and interfacing of input and output devices in digital systems. Hence, the working of a basic digital calculator was verified.